Group #4 Final Report

Enoch Yin, Jiarui Liang

1. Introduction

Skin cancer is one of the most prevalent cancers worldwide, with melanoma posing a particularly high risk due to its rapid progression and high mortality rate if not detected early. Accurate and timely diagnosis is therefore critical for improving patient outcomes. However, traditional dermatological diagnosis relies heavily on visual inspection and dermoscopic analysis, which can be subjective and dependent on clinician expertise.

Recent advances in deep learning have demonstrated strong potential in medical image analysis. Convolutional neural networks (CNNs) have achieved high performance in image classification tasks, while more recently, Vision Transformers (ViT) have shown competitive or superior performance by capturing global contextual relationships through self-attention mechanisms.

In this project, we develop DermAI, a skin cancer risk classification system that integrates deep learning models with clinically meaningful decision support. The system performs 10-class skin condition classification and further converts model outputs into a continuous cancer risk score (0–100) based on clinically defined malignancy weights.

To address clinical priorities, we introduce a clinical mode that adjusts decision thresholds using ROC-based optimization, prioritizing sensitivity for high-risk conditions such as melanoma. In addition, the system includes attention-based visualization and an interactive interface, aiming to bridge the gap between machine learning predictions and real-world clinical applications.

2. Description of the Dataset

To improve both generalization performance and clinical relevance, we constructed a multi-source dataset by integrating several publicly available skin image datasets. The primary dataset used in this study is HAM10000, which contains dermoscopic images of common pigmented skin lesions. While this dataset provides high-quality labeled dermoscopic images, it is limited in terms of class diversity and representation of non-cancer conditions.

To address these limitations, we incorporated supplementary datasets from Kaggle and ISIC 2020. The Kaggle datasets were used to introduce additional non-cancer skin conditions, including fungal infections such as tinea, as well as normal skin images covering different skin types such as dry, oily, and healthy skin. Furthermore, melanoma samples from the ISIC 2020 dataset were added to enhance the representation of high-risk classes and mitigate class imbalance issues.

All images from different sources were mapped into a unified 10-class classification framework, including both malignant and benign conditions as well as non-disease classes. Each class was assigned a predefined index and an associated clinical risk weight, which was later used to compute a continuous cancer risk score. Specifically, the risk score is defined as a weighted sum of predicted class probabilities, given by

$$\text{Risk Score} = \sum_{i=1}^{K} P(y_i \mid x) \cdot w_i \times 100,$$

where $P(y_i \mid x)$represents the predicted probability for class i, and $w_i$denotes the corresponding clinical risk weight.

To prevent data leakage and ensure reliable evaluation, we adopted a patient-level splitting strategy for the HAM10000 dataset using GroupShuffleSplit. This ensures that images from the same patient or lesion do not appear in both training and testing sets. The dataset was divided into training, validation, and test sets with proportions of approximately 75%, 15%, and 10%, respectively. For the supplementary datasets, stratified random splitting was applied to preserve class distribution across splits.

All images were processed using a standardized preprocessing pipeline to ensure consistency across datasets. Each image was first converted to RGB format and center-cropped to remove irrelevant background regions. Contrast Limited Adaptive Histogram Equalization (CLAHE) was then applied in the LAB color space to enhance lesion contrast and improve feature visibility. Finally, images were resized to 224×224 pixels to match the input requirements of the neural network models.

To improve robustness and reduce overfitting, extensive data augmentation techniques were applied during training. These include geometric transformations such as rotation and flipping, spatial distortions, noise injection, and color perturbations. Additionally, random occlusion techniques such as coarse dropout were used to encourage the model to learn more generalized features.

Due to significant class imbalance in the dataset, particularly for rare and clinically important classes such as melanoma, we employed a WeightedRandomSampler during training to generate balanced mini-batches. In addition, class weights were incorporated into the loss function to further emphasize underrepresented and high-risk categories.

3. Deep Learning Network and Training Algorithm

To evaluate model performance, we implemented two deep learning architectures: EfficientNet-B3 as a convolutional neural network baseline, and Vision Transformer (ViT) as the primary model. EfficientNet-B3 is a CNN architecture that scales network depth, width, and resolution in

a principled manner, allowing for efficient feature extraction with relatively low computational cost. The model was initialized with ImageNet-pretrained weights, and its final classification layer was replaced with a fully connected layer corresponding to the 10 target classes.

The main model used in this study is the Vision Transformer (ViT), which processes images as sequences of patches rather than grids of pixels. Each input image is divided into fixed-size patches of 16×16, and each patch is linearly embedded into a vector representation. A learnable classification token is appended to the sequence, and positional embeddings are added to preserve spatial information. The input to the transformer encoder can be written as

$$Z_0 = [x_{class}; x_p^1; x_p^2; \dots ; x_p^N] + E_{pos},$$

where x_p^i denotes the embedding of the i-th patch and E_{pos} represents positional embeddings.

The transformer encoder applies multi-head self-attention to model relationships between patches. The attention mechanism is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

where Q, K, and V denote query, key, and value matrices. This mechanism enables the model to capture global dependencies across the image, which is particularly useful for medical images where spatial context is important.

The models were trained using a weighted cross-entropy loss function to address class imbalance and emphasize clinically important categories. The loss is defined as

$$\mathcal{L} = -\sum_{i=1}^{K} w_i \, y_i \log \, \hat{y}_i,$$

where w_i represents class-specific weights, y_i is the ground truth label, and $\hat{y}_i$ is the predicted probability. Label smoothing was applied to improve generalization and reduce overconfidence by distributing a small portion of probability mass across all classes.

For the Vision Transformer, we adopted a two-phase fine-tuning strategy. In the first phase, the transformer encoder was frozen and only the classification head was trained for several epochs with a relatively large learning rate. In the second phase, all layers were unfrozen and the entire model was fine-tuned with a smaller learning rate. This approach helps stabilize training and preserves useful pretrained representations while adapting the model to the target task.

4. Experimental Setup

The models were trained using mini-batch optimization with the AdamW optimizer. Mini-batch size was selected based on available GPU memory, typically set to 32, which provides a balance between computational efficiency and gradient stability. Learning rates were determined empirically, with a higher learning rate used during the initial classifier training phase and a lower learning rate applied during full fine-tuning. A cosine annealing learning rate scheduler was employed to gradually decrease the learning rate during training, following the function

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})\left(1 + \cos\left(\frac{t}{T}\pi\right)\right).$$

Model performance was evaluated using multiple metrics, including accuracy, macro F1 score, and weighted F1 score. Among these, macro F1 score was selected as the primary evaluation metric due to the strong class imbalance in the dataset, as it assigns equal importance to all classes regardless of sample size.

To prevent overfitting, several regularization techniques were applied, including extensive data augmentation, weight decay, and label smoothing. Model checkpoints were selected based on validation macro F1 performance to ensure generalization to unseen data.

In addition to standard evaluation metrics, we incorporated clinically motivated evaluation strategies. Specifically, we evaluated high-risk recall across critical classes and performed ROC analysis to determine optimal decision thresholds. The thresholds were selected using Youden's J statistic, defined as

$$J = \mathrm{TPR} - \mathrm{FPR},$$

which balances sensitivity and specificity. This approach enables the model to prioritize detection of high-risk conditions, reflecting real-world clinical decision-making requirements.

5. Results

We evaluated the performance of both the Vision Transformer (ViT-base-patch16-224) and EfficientNet-B3 models on the test set using multiple metrics, including accuracy, macro F1 score, weighted F1 score, and recall for clinically high-risk classes. The results demonstrate that the ViT model consistently outperforms the EfficientNet baseline across most evaluation criteria.

In terms of overall classification performance, the ViT model achieved an accuracy of 76% and a macro F1 score of 0.6678, while EfficientNet-B3 obtained lower performance. The improvement

in macro F1 score indicates that the ViT model provides better balanced performance across all classes, particularly in the presence of class imbalance.

To better reflect clinical priorities, we introduced a high-risk recall metric, defined as the average recall across melanoma, basal cell carcinoma, and squamous cell carcinoma. In standard inference mode, the ViT model achieved a high-risk recall of 68%, compared to 58% for EfficientNet. This result suggests that the transformer-based architecture is more effective at capturing features relevant to malignant conditions.

To further improve sensitivity for critical cases, we applied the proposed clinical mode, which adjusts prediction thresholds based on ROC analysis. Under this setting, the ViT model achieved a high-risk recall of 81%, while EfficientNet reached 76%. This improvement demonstrates the effectiveness of threshold optimization in reducing false negatives for high-risk classes. However, this gain in recall is accompanied by a moderate decrease in overall accuracy and macro F1 score, reflecting a trade-off between sensitivity and specificity. Such a trade-off is consistent with clinical decision-making, where missing a malignant case is significantly more critical than generating false positives.

To analyze model behavior in greater detail, we examined the confusion matrix and per-class F1 scores. The confusion matrix reveals that the model performs well on well-represented and visually distinct classes, such as healthy skin and melanocytic nevi, where F1 scores exceed 0.8. Squamous cell carcinoma also shows strong performance, with an F1 score of 0.90 and recall of 0.81, indicating that the model is capable of identifying certain high-risk conditions reliably.

However, performance on melanoma and actinic keratosis remains relatively low. The melanoma class achieves an F1 score of 0.37 and recall of 0.40, while actinic keratosis has an F1 score of 0.34 and recall of 0.25. These results highlight the challenges associated with class imbalance, limited training samples, and visual similarity between different skin conditions. In particular, melanoma often shares visual characteristics with benign lesions, making it difficult for the model to distinguish reliably.

Further insight is provided by the ROC analysis for high-risk classes. The area under the ROC curve (AUC) indicates that the model still maintains reasonable discriminative ability for these conditions, even when classification performance is limited. The optimal thresholds derived using Youden's J statistic demonstrate that adjusting decision boundaries can significantly improve sensitivity, reinforcing the importance of probability-based evaluation rather than relying solely on argmax predictions.

Overall, the results show that the Vision Transformer model provides improved performance compared to CNN-based baselines, particularly in terms of balanced classification and high-risk detection. Moreover, the introduction of clinical mode and risk-based evaluation highlights the importance of aligning machine learning performance metrics with real-world medical priorities.

6. Summary and Conclusions

In this project, we developed DermAI, a deep learning-based system for skin disease classification and cancer risk assessment. By integrating multiple datasets, including HAM10000, Kaggle, and ISIC 2020, we constructed a diverse dataset covering both malignant and benign conditions, as well as normal skin images. We further designed a unified classification framework and introduced clinically motivated risk weights to enable continuous risk scoring.

We implemented and compared two deep learning architectures: EfficientNet-B3 as a convolutional baseline and Vision Transformer (ViT) as the primary model. Experimental results show that the ViT model outperforms the CNN baseline in terms of macro F1 score and high-risk recall, demonstrating its ability to capture global image features and improve classification performance in imbalanced medical datasets.

A key contribution of this work is the integration of clinical decision support mechanisms. The proposed risk scoring system translates model outputs into interpretable risk levels, while the clinical mode improves sensitivity for high-risk conditions by adjusting decision thresholds. These features make the system more aligned with real-world clinical requirements compared to traditional classification models.

Despite these promising results, several limitations remain. First, the dataset includes both dermoscopic and non-dermoscopic images, leading to domain mismatch that may affect model generalization. Second, class imbalance remains a challenge, particularly for melanoma and actinic keratosis, resulting in lower performance for these categories. Third, although attention-based visualization improves interpretability, it does not fully guarantee clinical reliability.

Future work can address these limitations in several ways. Domain adaptation techniques can be applied to reduce the gap between dermoscopic and clinical images. Additional high-quality labeled data, especially for rare and high-risk conditions, can further improve model performance. More advanced architectures or hybrid models combining CNN and transformer features may also enhance classification accuracy. Finally, integrating clinical metadata and patient information could provide a more comprehensive diagnostic framework.

Overall, this project demonstrates the potential of combining deep learning models with clinically informed design to create a more practical and interpretable skin cancer risk assessment system.

7. References

- HAM10000 Dataset: https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000
- ISIC 2020 Dataset: https://www.isic-archive.com
- Hugging Face ViT Model: https://huggingface.co/google/vit-base-patch16-224
- PyTorch Documentation: https://pytorch.org
- Albumentations: https://albumentations.ai
- Streamlit: https://streamlit.io
- EfficientNet Paper: Tan, M. and Le, Q., 2019
- Vision Transformer Paper: Dosovitskiy et al., 2021

8. Appendix

The full source code is publicly available in the project repository:

GitHub Repository:
https://github.com/zyinaa/Deep-Learning-Final-Project-Group-4